Course Outcome
After Learning the Course the students shall be able to:
1.Model and simulate cyber-physical systems using appropriate tools and techniques. 2.Analyze interactions between embedded software and physical processes. 3.Design CPS architectures for safety, reliability, and real-time constraints. 4.Apply CPS concepts to domains such as robotics, IoT, and intelligent transportation systems.

Parul University

Subject Syllabus

303105376 - Information and Network Security Laboratory

Course: BTech **Semester:** 6

Prerequisite: Basic knowledge of computer networks, cryptography, operating systems, programming, data structures, and cybersecurity fundamentals is required.

Rationale : This course introduces the fundamental principles of cryptography and its applications in the network security domain as well as the software development domain. This subject covers various important topics concerned with information security like symmetric and asymmetric cryptography, hashing, message and user authentication, digital signatures, key distribution, and an overview of the malware technologies. The subject also covers the applications of all of these in real- life situations.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Outcome
After Learning the Course the students shall be able to:
1. Analyze the trade-offs between security and complexity in the context of classical ciphers. 2. Apply the principles behind symmetric and asymmetric cryptography. 3. Demonstrate proficiency in hashing algorithms. 4. Apply message authentication techniques and their principles of digital signature and digital certificates. 5. Implement the various key management and remote authentication mechanisms.

List of Practical

1.	**Implement Caesar cipher encryption-decryption.**
2.	**Implement Monoalphabetic cipher encryption-decryption.**
3.	**Implement Playfair cipher encryption-decryption.**
4.	**Implement Polyalphabetic cipher encryption-decryption.**
5.	**Implement Hill cipher encryption-decryption.**
6.	**Implement Simple Transposition encryption-decryption.**
7.	**Implement One time pad encryption-decryption.**
8.	**Implement Diffie-Hellman Key exchange Method.**
9.	**Implement RSA encryption-decryption algorithm.**
10.	**Demonstrate working of Digital Signature using Cryptool.**